1.Create a table listing 5 key cybersecurity terms (such as threat, vulnerability, risk, asset, attack vector) and write a one-line definition for each in your own words.

Ans :-

No.	Cybersecurity Term	One-Line Definition
1	Threat	A threat is anything that can potentially cause harm to a computer system, network, or data.
2	Vulnerability	A vulnerability is a weakness in a system that an attacker can exploit.
3	Risk	Risk is the possibility of damage or loss when a threat takes advantage of a vulnerability.
4	Asset	An asset is anything valuable that needs protection, such as data, devices, software, or accounts.
5	Attack Vector	An attack vector is the method or path an attacker uses to gain unauthorized access to a system.

2.Research and find a real-world news article from the last 2 years about a major cyberattack on an Indian company or app (for example, a breach involving Zomato, Paytm, or IRCTC). Summarize what happened and identify one threat, one vulnerability, and one risk mentioned in the incident.

Ans :-

In July 2024, cybersecurity researcher Nilabh Rajpoot discovered a severe flaw in the IRCTC insurance portal, which is managed by a third-party provider, United India Insurance Co. Ltd. After booking a train ticket and opting for travel insurance, he found that by entering random PNR numbers and fictitious mobile numbers, the portal would reveal a passenger's full travel details—including journey date, train number, berth/seat, email, and mobile phone number—along with their insurance policy information.

More alarmingly, the portal allowed anyone to modify the nominee details in the insurance policy without requiring any OTP or security question. The researcher reported the issue to CERT-In on July 23, 2024, and by July 30, the vulnerability had been confirmed as fixed by the concerned organization.

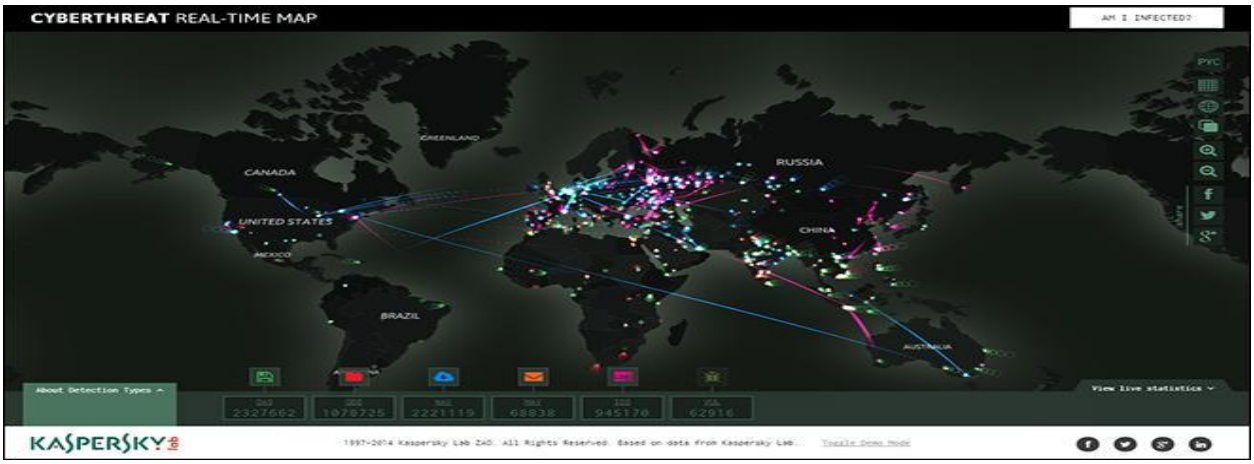
- **Threat**
  The primary threat was unauthorized access and data manipulation. Malicious actors could exploit the portal to gain unauthorized access to passengers' personal and travel information and, critically, alter nominee details in insurance policies. This posed a direct threat of fraud and data theft.

- **Vulnerability :-**
  The core vulnerability was a broken access control mechanism, specifically the absence of OTP verification for accessing and modifying sensitive data. The portal failed to properly authenticate users before displaying passenger details and allowing changes to nominee information, making it trivial for anyone with a PNR number to access private data.

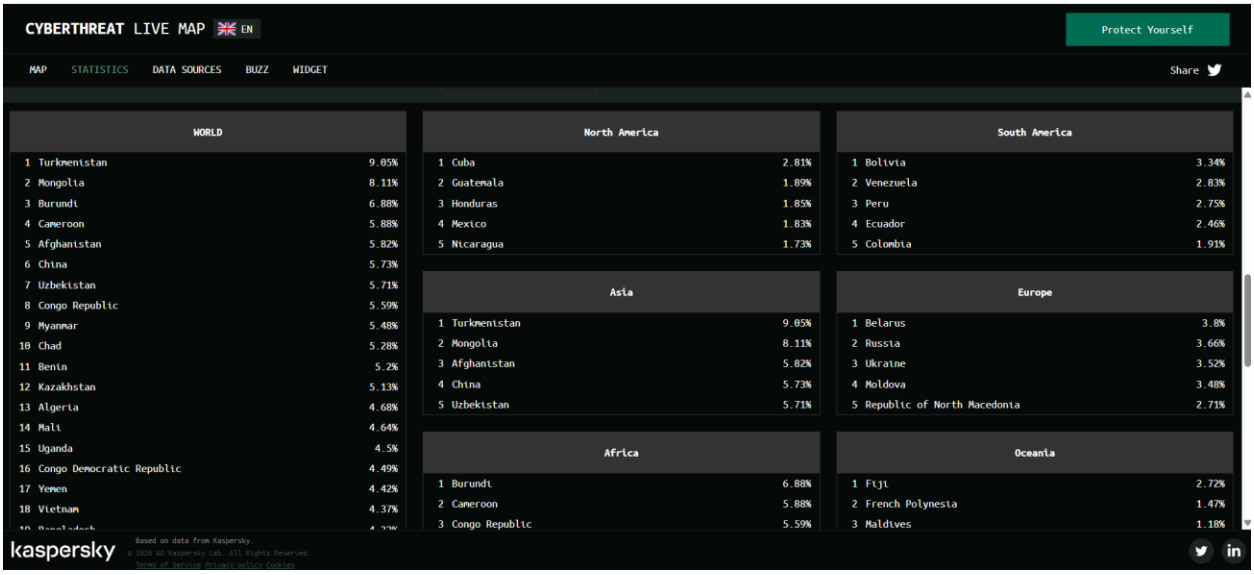
- **Risk :-**
  The incident created a significant risk of exposure of sensitive passenger information. The leaked data included journey details and contact information, which could be used for targeted phishing, identity theft, or other fraudulent activities. Furthermore, the ability to change nominee details introduced a risk of financial fraud and manipulation of insurance policies.

3.Sign up for a free account on any online cybersecurity dashboard or threat intelligence tool (such as https://www.cybermap.kaspersky.com/ or https://threatmap.checkpoint.com/). Take a screenshot of the dashboard and list 3 types of information or metrics you see displayed.

Ans :-







No.	Information / Metric	What it Shows
1	**Detections per Second**	Shows the approximate number of cyberthreat detections being recorded per second
2	**Most-Attacked Country**	Shows countries experiencing high numbers of detected cyberthreats.
3	**Threat Detection Types**	Shows categories such as On-Access Scan (OAS), On-Demand Scan (ODS), Mail Anti-Virus (MAV), Web Anti-Virus (WAV), Intrusion Detection (IDS), vulnerabilities (VUL), and ransomware (RMW).

**4.Write a simple phishing email template targeting users of a popular app (for example, 'Your Paytm account is suspended, click here to verify'). Highlight the suspicious elements that make it a phishing attempt.

Hint: Think about fake links, urgent language, or requests for sensitive info.**

Ans :-

From: Paytm Security `<security@example-fake.com>`

Subject: ⚠️ URGENT: Your Paytm Account Has Been Suspended

Dear Paytm User,

We detected on your account. Your Paytm account has been temporarily suspended.

To avoid permanent suspension, you must verify your account within **24 hours**.

👉 **Click here to verify your account:**
`https://paytm-security-verify.example.com`

Please enter your password, OTP, and other account details to complete the verification.

If you do not complete verification, your account may be permanently blocked.

Thank you,
Paytm Security Team

5.Use ChatGPT or another AI tool to generate a simulated phishing email for a fictional food delivery app (like 'FoodiePal'). Copy the AI-generated email and explain which parts are likely to trick users and why.

Ans :-

From: FoodiePal Support `<support@foodiepal-refunds.example.com>`
Reply-To: foodiepal.helpdesk@example.net
Subject: ⚠ Action required: Your ₹500 FoodiePal refund is on hold

Dear Valued Customer,

We tried to process a refund of **₹500** for your recent order **#FP-48291**, but your payment verification failed.

Your FoodiePal account will be **temporarily locked within 24 hours** unless you confirm your details.

To receive your refund, click the secure link below and verify your:

- Registered mobile number
- UPI ID / card details
- OTP sent to your phone

Claim your refund now:
`https://foodiepal-refund.example.com/claim`

If you do not verify within 24 hours, the refund will be cancelled and your account may be suspended.

Thank you,
FoodiePal Support Team
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❖ Why This Email Is Likely to Trick Users :-

1. **Familiar brand name and tone**
   It uses “FoodiePal,” a fictional but believable food delivery brand. The email looks like a normal support message, with a formal signature and copyright line.

2. **Financial incentive — ₹500 refund**
   People are strongly motivated by money. A refund notification grabs attention and makes users want to click quickly.
3. **Fake personalization**
   It includes an order number (`#FP-48291`) and a specific amount (`₹500`). Even if these are invented, they make the email feel legitimate and tailored to the recipient.
4. **Suspicious link domain**
   The link goes to `foodiepal-refund.example.com`, not the official `foodiepal.com`. The extra words "refund" and "example" are designed to look related. Also, **HTTPS does not guarantee safety** — phishing sites can use HTTPS too.
5. **Mismatched reply-to address**
   The "From" address is `@foodiepal-refunds.example.com`, while replies go to `@example.net`. This mismatch is common in phishing.
6. **Mobile-friendly, short email**
   It is brief and easy to read on a phone. On mobile, users often cannot see the full URL before tapping, making the fake link more dangerous.
7. **Emotional triggers**
   It combines anxiety ("account locked") with excitement ("refund"). This emotional mix makes users less likely to verify through official channels.